The Journal of Open Source Software

Chemical Recommender System: Replacement Suggestions for Small Molecules

Parthiv A. Nair[1], **Matthew D. Witman**[1], **Matthew E. Humphries**[2], and **Richard A. Karnesky**[1]

1 Sandia National Laboratories, Livermore, United States of America **2** Sandia National Laboratories, Albuquerque, United States of America

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 07 January 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

The Chemical Recommender System (CRS) is a modular and extensible open source toolkit for identifying and scoring replacement candidates for targeted small molecules. Filtered by customizable criteria, the candidates are scored based on predicted similarity, toxicity, and synthetic accessibility. First, we implement a fast vector databasing solution that permits pairwise Tanimoto similarity comparison between a target molecule and O(100) million candidate SMILES strings in minutes using just a single-chip CPU resource. This creates a shortlist of structurally similar candidates that can be additionally filtered according to substructure match requirements, elemental exclusion, etc. Next, we utilize open-source toolkits (rdkit and quantitative structure-activity relationship (QSAR) models in the OPERA package) to predict additional structural, thermophysical, synthetic accessibility, and toxicity properties for computing the final replacement score. Most importantly, the CRS's modular design allows users to easily integrate their own predictive models via Docker containers and customize how final replacement scores are calculated for application specific down-selections. Finally, the CRS is wrapped into a fully functioning web application which supports interactive queries with replacement score re-weighting post-processing. PDF reports are automatically generated with each query and batch jobs may be submitted for a comprehensive larger report.

Statement of need

Small molecules frequently become 'at risk' or unavailable due to supply, legislative, or technical issues (e.g., the recent discontinuation of PFAS by 3M). In such cases, identifying replacements quickly becomes essential. The CRS has been created to provide a first step in the down-selection of possible replacements for a given target molecule.

Chemical recommender systems have been widely explored, such as in Recommender Systems for Organic Compounds (Hayashi et al., 2022), which used machine learning techniques for classification, and DeepChem (Ramsundar et al., 2017), which applies deep learning models to predict solubility and toxicity in drug discovery. The CRS rather focuses mainly on adaptability/flexibility by permitting facile integration of additional user-provided structure-property models from which similarity metrics can be calculated. Other similarity searches, such as those from PubChem and SigmAldrich, lack technical transparency, customizability, and extensibility, which the CRS offers in an open-source format. Moreover, the largest benefit of the CRS is its modular and extensible form, which allows users to add their own models through Docker containers, whose predictions can be integrated in the metrics used to score replacements.

Software Overview

Program Structure

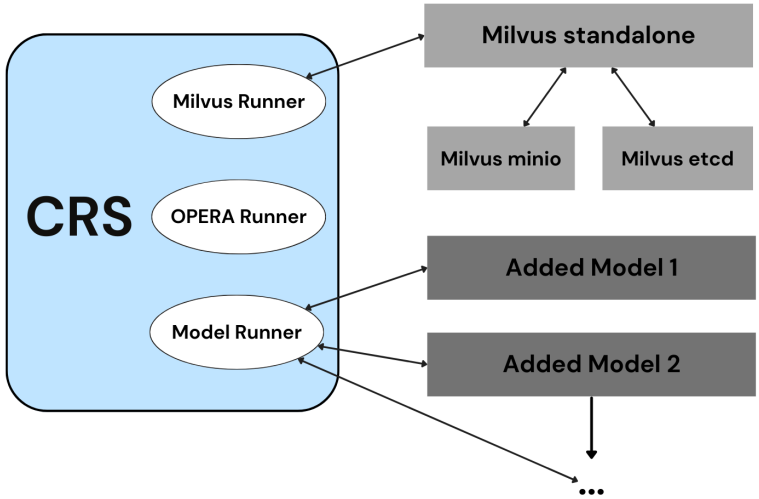

Figure 1: The CRS works as a series of containers. They all interact with one another stemming from the CRS using runners.

The CRS is a series of containers. Users can pull and initialize all necessary containers and dependencies with ease by using the docker-compose file provided in the CRS repository. The CRS is an overarching container that initializes runners who interact with the dependencies through port connections, through which necessary information is retrieved into the CRS for further processing and ultimately contributes to the final replacement scores (FRS).

Initial Structural Similarity (Vector Databasing)

In a one-time preprocessing step, the RDKit calculates the 2048-bit vector Morgan Fingerprint (Cereto-Massagué et al., 2015) representations of each SMILE for all PubChem (Kim et al., 2023) CIDs (available through PubChem FTP), i.e., the list of all possible replacements for any user provided query molecule. These CID-Fingerprint values are then stored into a vector database, Milvus (Wang et al., 2021), to enable fast structural similarity comparisons.

Milvus interacts with the CRS as three containers. The Milvus standalone container interacts directly with the CRS for API requests. The Milvus etcd and Milvus minio containers provide support to the standalone by maintaining vector information with their indexed metadata. This database was filled by the CRS developers and is included in the container via the provided docker compose. This is implemented by using shared volumes that are stored in the CRS and flow over into the Milvus containers.

At runtime, a fingerprint is computed for the queried molecule, which can be any entry in PubChem, referenced by name, CID, or SMILES. Alternatively, users can also provide a query for any hypothetical SMILES that may not exist in PubChem. The computed fingerprint is compared against all other candidates using Tanimoto similarity Equation 1, which returns a float from 0 to 1, providing top candidates to be shortlisted for further comparison.

$$Tani(F_i, F_j) = \frac{F_i \cdot F_j}{\sum_k F_{i_k} + \sum_k F_{j_k} - F_i \cdot F_j} \tag{1}$$

DRAFT

This Milvus vector database implementation results in ~60x speed ups as compared to a traditional max-heap implementation, when must be computed for a possible replacement list that is the size of the full PubChem database. Finally, a user-specified number of remaining candidates with the highest Tanimoto similarity are retained for further processing.

Additional Filter Criteria

A series of user-specified filters can additionally be applied to exclude possible replacements. For example, users can specify which elements are permitted in possible replacement candidates. Users can additionally filter candidates by requiring the existence of specific SMARTs substructures with a specific number of occurrences.

OPERA Predictions

OPERA (Mansouri et al., 2018) is an open-source suite of QSAR models that provide predictions for thermophysical, structural, and toxicity properties of molecular compounds. After the structurally similar replacement shortlist is created, OPERA's models predict these properties for each replacement candidate and the query molecule.

By default, OPERA computes additional structural properties (molecular weight, number of rings, number of Lipinski failures, number of carbon atoms, and topological polar surface area) and thermophysical properties (melting point, boiling point, logP, Vapor Pressure, and Henry's Law Constants). These represent single-valued scalars for which each candidate's property y is compared against the query's property x, in order to assign similarity scores (C_1: Structural Similarity, C_2: Thermophysical Similarity, C_3: Molecular Weight Similarity).

Toxicity endpoints are also predicted by OPERA and factored into the FRS. These calculated endpoints are Log BCF (fish bioconcentration factor), CATMoS (Collaborative Acute Toxicity Modeling Suite) for EPA categories, and LD50, which are combined as a product into a single Toxicity Score, C_4.

Synthetic Accessibility

Synthetic accessibility Score (Skoraczyński et al., 2023), C_5, ranges from 1 (easy) to 10 (difficult) and describes synthesis difficulty as calculated using a fragment contribution model trained on PubChem fingerprints.

Additional Models and Weightage

Users can add their own models to the CRS as new conatiners, so long as they take a SMILES as an input and return a numerical value as an output. The CRS then evaluates similarity of the model predictions for each candidate to that of the query molecule. Each nth model contributes its own category to FRS, as C_{5+n}.

Finally, each scoring category, C_i, is assigned a corresponding weight W_i, and FRS $= f(\{C_i\}, \{W_i\})$ is calculated. The functional form of f and values of $\{W_i\}$ can be user-customized for desired functionality. Candidates are then ranked by FRS and returned with data and figures via webapp or PDF report.

The Journal of Open Source Software

Figure 2: Visualization of CRS output from the web app searching interface, including the query molecule, possible replacements, and (C_1, C_2, C_3, C_4, C_5) for top CRS candidates.

Code Availability

The CRS source code is available for free on Github under the BSD-3-Clause license (https://github.com/sandialabs/chemical-recommender-system), and can be used to install and run the CRS.

Acknowledgments

Sandia National Laboratories is a multi-mission laboratory managed and oper- ated by National Technology & Engineering Solutions of Sandia, LLC (NTESS), a wholly owned subsidiary of Honeywell International Inc., for the U.S. Department of Energy's National Nuclear Security Administration (DOE/NNSA) under contract DE-NA0003525. This written work is authored by an employee of NTESS. The employee, not NTESS, owns the right, title and interest in and to the written work and is responsible for its contents. Any subjective views or opinions that might be expressed in the written work do not necessarily represent the views of the U.S. Government. The publisher acknowledges that the U.S. Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this written work or allow others to do so, for U.S. Government purposes. The DOE will provide public access to results of federally sponsored research in accordance with the DOE Public Access Plan.

References

Cereto-Massagué, A., Ojeda, M. J., Valls, C., Mulero, M., Garcia-Vallvé, S., & Pujadas, G. (2015). Molecular fingerprint similarity search in virtual screening. *Methods*, *71*, 58–63. https://doi.org/10.1016/j.ymeth.2014.08.005

Hayashi, H., Seko, A., & Tanaka, I. (2022). Recommender system for discovery of inorganic compounds. In *npj Computational Materials* (Vol. 8). https://doi.org/10.1038/s41524-022-00899-0

Kim, S., Chen, J., Cheng, T., & al., et. (2023). PubChem 2023 update. *Nucleic Acids Research*, *51*(D1), D1373–D1380. https://doi.org/10.1093/nar/gkac956

Mansouri, K., Grulke, C. M., Judson, R. S., & al., et. (2018). OPERA models for predicting physicochemical properties and environmental fate endpoints. *Journal of Cheminformatics*, *10*, 10. https://doi.org/10.1186/s13321-018-0263-1

Ramsundar, B. K., Eastman, P., Walters, P., John, R. St., Huang, X., & Pande, V. S. (2017). DeepChem: A python library for deep learning in drug discovery. In *arXiv preprint arXiv:1703.00067*. https://doi.org/10.48550/arXiv.1703.00067

Skoraczyński, G., Kitlas, M., Miasojedow, B., & al., et. (2023). Critical assessment of synthetic accessibility scores in computer-assisted synthesis planning. *Journal of Cheminformatics*, *15*, 6. https://doi.org/10.1186/s13321-023-00678-z

Wang, J., Yi, X., Guo, R., Jin, H., Xu, P., Li, S., Wang, X., Guo, X., Li, C., Xu, X., Yu, K., Yuan, Y., Zou, Y., Long, J., Cai, Y., Li, Z., Zhang, Z., Mo, Y., Gu, J., … Xie, C. (2021). *Milvus: A purpose-built vector data management system*. 2614–2627. https://doi.org/10.1145/3448016.3457550

DRAFT